

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - II

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Class :SYIT	Batch : 01
Date of Assignment :22/07/26	Date/Time of Submission : 22/07/2026

Questions:

2a. Write a python code to create nested tuples.

INPUT:

```
student=("sejal",6, ("BSCIT", "Sem3"), 90
)
print("name: ",student[0])
print("Rollno : ",student[1])
print("Department: ",student[2][0])
print("Semester: ",student[2][1])
print("percentage: ",student[3])

print("S006 Sejal Deshmukh")
```

OUTPUT:

```
= RESTART: C:/Users/itlab/OneDrive
prac2.py
name:  sejal
Rollno :  6
Department:  BSCIT
Semester:  Sem3
percentage:  90
S006 Sejal Deshmukh
>
```

2b. Write a python code to sort the nested tuple using sorted () function.

INPUT:

```
data= [(1,4),(3,2),(5,1)]
sort_data= sorted(data, key= lambda x :x[1])
print(sort_data)
print("S006 Sejal Deshmukh")
```

OUTPUT:

```
= RESTART: C:/Users/itlab/O
prac2.py
[(5, 1), (3, 2), (1, 4)]
S006 Sejal Deshmukh
|
```

2c. Write a python code to copy or clone list.

INPUT:

```
list1=(1,2,3,4,5)
list2=0

list2=list1

print(list2)
print("S006 Sejal Deshmukh")
```

OUTPUT:

```
prac2.py
(1, 2, 3, 4, 5)
S006 Sejal Deshmukh
```

2d. Write a python code to check immutability property of python tuples.

INPUT:

```
student=("sejal",6, ("BSCIT", "Sem3"), 90
)
student[1]=34

print("S006 Sejal Deshmukh")
```

OUTPUT:

```
>> = RESTART: C:/Users/itlab/OneDrive/Desktop/S006 Data Structure/PDS  
prac2.py  
Traceback (most recent call last):  
  File "C:/Users/itlab/OneDrive/Desktop/S006 Data Structure/PDSpra  
c2.py", line 3, in <module>  
    student[1]=34  
TypeError: 'tuple' object does not support item assignment  
>> |
```

Ln: 81 Col: 0

Curiosity Questions:

Q1. Define a tuple. How is it different from a list? Give one example.

Q2. Write a Python program to create the following tuples:

- a) An empty tuple
- b) A tuple with integers only
- c) A tuple with mixed data types

d) A nested tuple (tuple inside another tuple)

Q3. Write a program to access elements of a tuple using indexing and slicing.

Q4. Create a tuple and find the following:

- Length of the tuple
- Maximum and minimum value (if all are numbers)
- Count of a specific item

Q5. Write a program to convert a list into a tuple.

Q6. Write a program to sort a tuple of numbers in ascending order.

Q7. Write a program to show that tuples are immutable (cannot be changed).

Q8. Write a program to clone (copy) a list and print both original and copied lists.

Q9. Write the output of the following code:

```
t = (1, 2, 3, 4, 5)
```

```
print(t[1:4])
```

Q10. Create a tuple of student marks: (78, 85, 92, 88, 76). Write a program to:

- Find average marks
- Display marks above 80